Upliance AI: Hardware-Aware Generative Recipe & Personalization Engine

System Architecture & Compiler Specifications for the upliance Smart Appliance

Problem

Context & Physical Hardware Constraints

The upliance smart cooking hardware operates via precision physical actuators—motorized stirring blades, induction heating elements, and integrated weighing sensors. However, the appliance initially relied on static, hardcoded recipe programs that could not accommodate real-world kitchen variables: dietary requirements (keto, vegan, Jain), missing pantry items, custom portion sizes, or personal spice tolerances. Generic conversational LLMs (such as standard chatbots) cannot safely solve this, as unconstrained text generation cannot safely translate into the strict thermal limits, blade speeds, and timed physical steps required by physical kitchen hardware.

Solution

Engineered upliance AI, an intelligent, hardware-constrained recipe personalization and compiler engine that translates natural language culinary requests into safe, machine-executable cooking workflows:

- **Semantic Recipe Discovery (AI Vector Search & Retrieval):** Powers dynamic recipe discovery across the verified upliance recipe database using semantic vector embeddings (`pgvector`). Ingests user dietary profiles, available pantry ingredients, and time constraints to match and retrieve optimal kitchen-tested base recipes based on culinary meaning and flavor profiles rather than rigid keywords.
- **Context-Aware Recipe Adaptation Engine:** Developed an intelligent culinary reasoning pipeline using a self-hosted private AI model (Qwen LLM on private vLLM infrastructure). When a user requests on-the-fly modifications (e.g., *"make this low-carb, swap paneer for tofu, and adjust for 4 people"*), upliance AI dynamically recalculates ingredient proportions, liquid-to-solid moisture balance, and cooking duration.
- **Deterministic Machine Instruction Compiler:** Enforces strict data-validation rules (via Zod schemas) to compile natural language cooking steps into exact, deterministic machine controls:
  - **Motor Speeds:** Stirring RPM, reverse blade mode (stirring with the blunt edge to gently mix without shredding delicate ingredients like paneer or tofu), and pulse chopping cycles.
  - **Multi-Stage Induction Heat Curves:** Gradual, controlled temperature ramps and timed hold intervals to prevent burning or scorching.
  - **Integrated Scale Calibration:** Step-by-step weight prompts with automated tare calibration (automatically zeroing the scale as ingredients are added).
  - **Hardware Safety Envelopes:** Enforced physical guardrails (scorch prevention, dry-run cutoffs, and boil-over mitigation).
- **Real-Time Appliance Synchronization:** Connects upliance AI to the appliance's touchscreen operating system via lightweight real-time communication protocols (WebSockets), delivering live interactive cooking guidance, automated appliance actuator control, and responsive feedback to user taste adjustments.

Architecture & Execution Flowchart

The diagram below illustrates the complete execution pipeline—from user natural language requests, through the upliance AI compilation engine, to structured machine-code parameters controlling physical hardware actuators:

HOW THE PLATFORM WORKS, END TO END — Hardware-Aware Generative Recipe Compilation

1 INPUT • NATURAL LANGUAGE REQUEST — Spoken or Touchscreen Ingress

The user asks for something, in plain words

"Make this less spicy, swap the paneer for tofu, and cook it for 4 people."

Typed or spoken directly into the appliance's touchscreen — zero manual menus, zero rigid preset programs.

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2 UPLIANCE AI ENGINE • UNDERSTAND • ADAPT • COMPILE — HARDWARE-CONSTRAINED REASONING

1 Find the best-matching tested recipe

Searches upliance's library of kitchen-tested recipes by culinary meaning rather than rigid keywords, selecting the closest match to the user's diet, pantry inventory, and available cooking time.

`PostgreSQL + pgvector (HNSW semantic vector search)`

2 Rework the recipe around the request

Recalculates ingredient quantities, liquid-to-solid moisture balance, cooking times, and heat levels — ensuring tofu for 4 people is cooked properly, not just relabelled.

`Self-Hosted Private AI (Qwen LLM on vLLM infrastructure)`

3 Convert every step into exact machine settings

Each cooking step is rewritten as precise numbers the hardware can safely execute. Anything that fails validation or falls outside safe limits is rejected before it reaches the appliance.

BLADE ACTUATOR	INDUCTION HEAT	SCALE SENSOR
`Speed 2 RPM • Reverse gentle stir`	`110 °C • Hold 4 min curve`	`Prompt: Add 200 g diced tofu`

HARDWARE SAFETY ENVELOPE: Scorch prevention, boil-over thermal ceilings, dry-run cut-offs, and lid-lock interlocks are validated on every instruction.

`Strict schema validation • Embedded firmware interlocks`

VALIDATED MACHINE-CODE JSON OVER WEBSOCKET

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3 OUTPUT • PHYSICAL APPLIANCE EXECUTES STEP BY STEP — Synchronized Actuation

Touchscreen UI

Shows the adapted recipe one step at a time, providing live visual guidance, countdowns, and on-the-fly taste adjustments.

Built-In Scale

Weighs dynamically as the user pours (e.g., *"Added 150 g of 200 g"*) with automated tare zeroing until the amount is exact.

Motor & Induction

Stirs and heats automatically at the exact compiled RPM, temperature curve, and duration compiled by the AI engine.

Hardware-Software Compilation & Safety Matrix

Architectural layer breakdown, core technologies, and operational responsibilities across the upliance smart cooking platform:

SYSTEM LAYER	SUBSYSTEM NAME	CORE TECHNOLOGY	OPERATIONAL SCOPE & RESPONSIBILITIES
Layer 1	**User Intent Ingestion**	Touchscreen / Voice NLP	Captures natural culinary substitutions, dietary preferences (keto, vegan, Jain), and portion scaling.
Layer 2A	**Semantic Discovery & RAG**	PostgreSQL + pgvector (HNSW)	Matches user taste preferences and pantry ingredients to certified, kitchen-tested base recipes via vector search.
Layer 2B	**Culinary Adaptation Engine**	Self-Hosted Qwen via vLLM	Recalibrates ingredient volume, liquid-to-solid moisture balance, cooking durations, and induction heat curves.
Layer 2C	**Deterministic Hardware Compiler**	Strict Schema Validation	Outputs validated machine code for blade stirring RPM, reverse blade modes, heat curves, and scale calibration.
Safety Envelope	**Hardware Guardrail Engine**	Embedded Firmware Interlocks	Enforces physical safety limits: burn prevention, boil-over thermal ceilings, dry-run heater cutoffs, and lid-lock triggers.
Hardware Layer	**upliance Smart Appliance**	Actuators & Physical Sensors	Automated motorized blade stirring, precision induction heating, live scale with tare calibration, and touchscreen UI.

Platform Engineering Highlights & Hardware Guardrails

Institutional Safety Standards for Kitchen Automation

- **Hardware-Aware Culinary Reasoning:** Unlike generic conversational chatbots that generate ungrounded recipes, upliance AI incorporates physical kitchen physics—calculating evaporation rates, induction thermal inertia, and blade shear resistance before generating instructions.
- **Reverse Blade Texture Preservation:** Automatically deploys reverse blade rotation (stirring with the blunt edge of the blade) during delicate simmering stages, preventing soft proteins like paneer, tofu, or vegetables from being pureed.
- **Fail-Safe Firmware Interlocks:** All compiled machine parameters pass through local microcontroller firmware checks. If sensors detect insufficient liquid (dry-run risk) or abnormal temperature spikes, the induction heater immediately cuts off independently of cloud software.
- **Closed-Loop Interactive Weighing:** Synchronizes recipe steps with integrated load-cell scales, pausing execution until the user reaches the exact ingredient weight target and automatically zeroing (tare calibration) for the next step.